1. Dataset
- Collection of data used for training
- Divided into:
  - Training data → used to learn
  - Testing data → used to check performance

2. Features and Labels
- Features (X) → input data
- Labels (Y) → correct output

Example:
- Input: Image of "5"
- Output: 5

3. Neural Network
A neural network is a model made of layers of neurons.
Structure:
- Input layer → takes data
- Hidden layers → learn patterns
- Output layer → gives result

4. Activation Function
Used to introduce learning ability.
ReLU (commonly used)

$$f(x) = \max(0, x)$$

- Converts negative values to zero
- Helps model learn faster

5. Output Activation
- Sigmoid → for binary classification
- Softmax → for multiclass classification

6. Loss Function
Measures error between:
- Actual output
- Predicted output

Examples:
- Binary crossentropy
- Categorical crossentropy

7. Optimizer
Updates model weights to reduce error
Example:
- RMSprop
- Adam

8. Training

```
model.fit(...)
```

Important terms:
- Epoch → number of times model sees data
- Batch size → number of samples processed at once

9. Evaluation

```
model.evaluate(...)
```

Gives:
- Loss
- Accuracy

10. Accuracy
- Shows how correct the model is

Example:
Accuracy = 95% → model is mostly correct

4. Full Flow of the Program
Data → Preprocessing → Neural Network → Training → Prediction → Evaluation

5. What you are learning from this practical
- How to build a neural network
- How to train it
- How to evaluate it
- How to classify data

This experiment uses a deep neural network to learn patterns from data and perform classification by predicting the correct class for given input.

Q) Image Classification using Convolutional Neural Network (CNN)
👉 Task:
Given an image (clothes), predict its class
Example: T-shirt/Dress/Bag/Shoe
👉 Dataset: **Fashion MNIST**

2. Type of Problem
Multiclass Classification
- Input → Image
- Output → One class out of 10

3. Dataset Concept
Fashion-MNIST contains:
- 60,000 training images
- 10,000 testing images
- Image size: **28 × 28 pixels**

Image Representation
Each image = matrix
- 28 rows × 28 columns
- Each value = pixel intensity (0 to 255)

4. Data Preprocessing

(1) Normalization

```
x_train = x_train / 255.0
```

Explanation:
- Convert pixel values from 0–255 → 0–1
- Helps faster learning

(2) Reshaping

```
x_train.reshape(-1, 28, 28, 1)
```

Explanation:
- Add channel dimension
- Required for CNN

5. Convolutional Neural Network (CNN)
Definition
CNN is a neural network specially designed for **image processing**.
👉 It automatically learns:
- edges
- shapes
- patterns

6. Important CNN Layers

(1) Convolution Layer (Conv2D)

```
Conv2D(32, (3,3), activation='relu')
```

Explanation:
- Applies filters on image
- Extracts features

Example:
- Detect edges
- Detect shapes

(2) Activation Function (ReLU)
Explanation:
- Converts negative values to 0
- Helps model learn faster

(3) Pooling Layer (MaxPooling)

```
MaxPooling2D((2,2))
```

Explanation:
- Reduces image size
- Keeps important features

(4) Dropout

```
Dropout(0.25)
```

Explanation:
- Randomly removes neurons
- Prevents overfitting

(5) Flatten Layer

```
Flatten()
```

Explanation:
- Converts 2D data → 1D vector
- Required before Dense layer

(6) Dense Layer

```
Dense(128)
```

Explanation:
- Performs final learning
- Combines features

(7) Output Layer

```
Dense(10)
```

Explanation:
- 10 neurons → 10 classes
- Each neuron represents one class

7. Output Function
Softmax
- Converts output into probabilities
- Sum = 1

Example:
[0.1, 0.2, 0.6, ...]
👉 Highest value = predicted class

8. Model Compilation

Loss Function

```
sparse_categorical_crossentropy
```

Explanation:
- Used for multiclass classification
- When labels are numbers (0–9)

Optimizer

```
adam
```

Explanation:
- Updates weights
- Improves learning

Metric

```
accuracy
```

Explanation:
- Measures correctness

9. Training Concepts

Epoch
- Number of times model sees full dataset

Batch Size
- Number of samples processed at once

10. Model Evaluation

```
model.evaluate(...)
```

Gives:
- Loss
- Accuracy

👉 Your result = 91%

11. Overfitting
Meaning:
Model learns too much from training data
→ performs poorly on new data
Solution:
- Dropout
- More data
- Regularization

12. Full Flow
Image → Normalize → CNN Layers → Feature Extraction → Classification → Output